

Smashifix: An Application for Badminton Posture Correction Using Deep Learning

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Abstract

Badminton is a high-speed racquet sport demanding precise biomechanics and rapid reflexes, where even minor deviations in stroke technique can lead to stagnated performance and chronic injury. Traditional coaching, reliant on subjective observation, is inconsistent, expensive, and inaccessible to amateur players; existing automated solutions either classify actions without corrective feedback or require expensive multi-sensor hardware. This paper presents “Smashifix,” an automated computer vision framework that addresses these gaps through a dual-pipeline architecture: a Pose-LSTM pipeline for real-time feedback (~ 50 ms latency) and an Attention-enhanced Temporal Convolutional Network (Attention-TCN) fusing skeletal data with visual context from a custom Residual-Shuffle Network (RSN) for forensic analysis ($\sim 85\%$ accuracy). A Kinematic Similarity Index (KSI) quantifies user performance against expert templates using Dynamic Time Warping (DTW), which a Natural Language Coach subsequently translates into personalized, skill-appropriate textual feedback. The framework classifies six distinct strokes and provides actionable biomechanical feedback using only a standard smartphone camera, enabling accessible, objective coaching for players at all levels.

Keywords: Markerless Pose Estimation, Attention-enhanced TCN, Residual-Shuffle Network, Kinematic Similarity Index, Biomechanical Feedback, Dynamic Time Warping, Natural Language Coach

1 Introduction

1.1 Background

The intersection of computer vision and artificial intelligence has fundamentally transformed the landscape of sports analytics, shifting the paradigm from subjective observation to data-driven decision-making [1]. Modern systems possess the capability to track intricate movement patterns, providing athletes and coaches with objective, quantitative data that was previously inaccessible [2]. In high-performance sports, even minor deviations in technique can significantly impact competitive outcomes, making robust tracking and re-identification essential [3]. Badminton, in particular, is a sport defined by rapid reflexes, explosive power, and precise body mechanics. It requires a complex interplay of footwork, stroke execution, and tactical positioning, where the shuttlecock can travel at speeds exceeding 400 km/h. Consequently, the ability to analyze movement in granular detail—down to the angle of the wrist or the rotation of the hip—is essential for player development and injury prevention.

Historically, biomechanical analysis was confined to laboratory settings equipped with marker-based motion capture systems. While highly accurate, these systems are intrusive, expensive, and restricted to controlled environments, making them unsuitable for on-court coaching. The advent of markerless pose estimation algorithms has opened new avenues for “in-the-wild” analysis, allowing for the extraction of kinematic data from standard video feeds. This democratization of technology presents a unique opportunity to bring professional-grade analytics to amateur and semi-professional players.

1.2 Current Limitations and System Objectives

Despite advancements in sports technology, badminton coaching largely remains a manual and subjective process. Players often face challenges with improper posture and technique, which not only hampers skill development but also significantly increases the risk of chronic injuries such as lateral epicondylitis (tennis elbow) and patellar tendinopathy. Current automated solutions often suffer from critical limitations that hinder their widespread adoption:

- **Hardware Dependency:** Many high-fidelity systems rely on expensive wearable sensors (IMUs) or multi-camera setups. These are impractical for widespread use due to their cost and setup complexity.
- **Lack of Corrective Feedback:** Most existing computer vision applications focus solely on action recognition (e.g., classifying a smash vs. a drop) or basic statistical tracking. They fail to provide the “how” and “why” of a mistake, lacking the corrective feedback necessary for motor learning.
- **Latency vs. Accuracy Trade-off:** Current approaches often fail to balance the need for real-time processing with the depth required for forensic biomechanical analysis. Real-time models are often too lightweight to capture subtle nuances, while high-accuracy models are too computationally intensive for live feedback.

- **Robustness of Spatial Tracking:** Dynamic bounding box methods based on joints frequently suffer from temporal jitter, occlusion-induced failures, and inconsistent scaling, leading to noisy downstream feature extraction that compromises forensic analysis.

To bridge these gaps, the system “Smashifix” is proposed as an accessible, single-camera coaching assistant. The primary objectives of this research are:

1. To develop a hardware-agnostic framework that provides immediate, actionable feedback on player posture using only a standard smartphone camera.
2. To integrate lightweight deep learning models (Pose-LSTM) for real-time cues with robust hybrid models (Attention-TCN with RSN) for detailed forensic analysis.
3. To mathematically quantify deviations from expert templates using an enhanced Kinematic Similarity Index (KSI), thereby reducing the risk of injury through automated, objective correction.
4. To overcome the limitations of dynamic bounding box tracking by employing a robust, configuration-driven static spatial cropping methodology that guarantees consistent player scaling.

The organization of this paper is as follows: Section 2 presents the Literature Survey, reviewing state-of-the-art methods in pose estimation and sports analytics, followed by a comparative analysis. Section 3 outlines the proposed system, detailing the novel dual-pipeline architecture. Section 4 provides an in-depth explanation of the methodology, including data preprocessing via deterministic spatial cropping, mathematical formulations for the KSI, and feature engineering. Section 5 presents the experimental setup, dataset description, evaluation metrics, and a comprehensive discussion of the results. Finally, Section 6 concludes the study and outlines directions for future research.

2 Literature Survey

This section reviews the foundational research and state-of-the-art methodologies that inform the development of the Smashifix framework. The literature is structured logically to cover the multifaceted domains of automated sports coaching. First, a survey of existing sports systems and specialized datasets is presented to establish the current data landscape. Following this, the evolution of human pose estimation architectures is explored, tracing the shift from early regression models to robust markerless tracking frameworks. The discussion then transitions to dynamic action recognition techniques tailored for high-velocity sports. To address the computational constraints of real-time on-court analysis, recent advancements in lightweight neural architectures and spatio-temporal attention mechanisms are examined. Furthermore, existing literature on biomechanical feedback and action quality assessment is reviewed to highlight the challenges of generating actionable coaching cues. Finally, a comparative analysis is provided to contextualize the proposed Smashifix system against contemporary solutions, underscoring the specific technological gaps this research aims to bridge.

2.1 Survey of Existing Systems and Datasets

The development of automated sports analysis relies heavily on the availability of high-quality, domain-specific data and robust architectures.

Wang et al. [4] detailed a structured approach to creating a tennis player dataset, capturing 2,000 annotated frames of key actions like forehands and serves using COCO-Annotator. This study emphasized the critical importance of diverse pose representation for training robust models capable of generalizing across different athletes. Expanding on racket sports data, Sozykin et al. [5] established comprehensive multi-camera datasets specifically aimed at tracking high-velocity sporting actions, paving the way for multi-view biomechanical assessments.

In the specific domain of badminton, Zhang et al. [6] developed a human-computer interactive teaching system using Part Affinity Fields (PAF) trained on a massive dataset of 35,000 movements. While their system was effective for basic movement tracking and hit-point detection, it primarily functioned as a retrieval system rather than a granular biomechanical correction tool. Further addressing badminton intricacies, Ghosh et al. [7] introduced towards-automated techniques for shuttlecock trajectory tracking, which, when combined with human pose, provides critical contextual data regarding the player’s positioning relative to the shuttle. Additionally, Gu et al. [8] demonstrated the utility of large-scale action recognition benchmarks, highlighting the necessity of combining diverse shot types to train models effectively for real-world scenarios.

A comprehensive overview of these Human Pose Estimation (HPE) techniques is provided by Dibenedetto et al. [9]. Their survey categorizes methods into skeleton-based, surface-based, and volume-based models, highlighting the evolution from regression-based methods to modern detection frameworks like MediaPipe and OpenPose. They note that while 2D estimation has matured, persistent challenges remain in handling self-occlusion and crowded scenes, which are common in doubles badminton matches. Furthermore, the use of Spatio-Temporal Graph Convolutional Networks (ST-GCN) has emerged as a powerful method for modeling the physical connectivity of the human skeleton in dynamic sports actions [10].

2.2 Pose Estimation Architectures

Accurate localization of joints is the cornerstone of biomechanical analysis. Cao et al. [11] established the viability of real-time 2D pose estimation with OpenPose, using Part Affinity Fields to associate keypoints in multi-person scenarios. This bottom-up approach was revolutionary but computationally expensive. Building on this, Lugaresi et al. [12] introduced MediaPipe, a modular framework optimized for on-device machine learning. Within this ecosystem, Bazarevsky et al. [13] proposed BlazePose, a lightweight detector-tracker architecture that predicts 33 body landmarks, making it ideal for real-time mobile applications where latency is a critical constraint.

Historically, Toshev and Szegeedy [14] pioneered the use of Deep Neural Networks (DNNs) for pose estimation with DeepPose, shifting the paradigm from manual feature engineering to direct regression of joint coordinates. For more advanced anatomical precision, Güler et al. [15] introduced DensePose, which maps 2D images to a 3D

surface model. This allows for the analysis of complex muscle activation patterns and torso rotations that simpler landmark-based models often miss. To address the inherent limitations of 2D data, Martinez et al. [16] proposed a simple yet effective baseline for lifting 2D joint locations to 3D space, which provides the depth necessary for identifying subtle stroke deviations.

2.3 Action Recognition in Sports

Recognizing dynamic sports actions requires modeling both spatial and temporal dependencies. In badminton, Tukino et al. [17] utilized Convolutional Neural Networks (CNNs) combined with BlazePose to classify strokes such as smashes and serves, achieving 80–90% accuracy. However, their system was limited to classification and lacked a feedback loop. Asriani et al. [18] achieved state-of-the-art results (98.68%) by proposing a hybrid model that fuses Vision Transformers (ViT) for global context with skeletal features extracted via MediaPipe.

In other sports, Siddiqui et al. [19] applied Random Forests and Long Short-Term Memory (LSTM) networks to classify cricket batting strokes, achieving 99.77% accuracy with kinematic features. To handle temporal dynamics, Donahue et al. [20] introduced Long-term Recurrent Convolutional Networks (LRCN), which combine CNNs for visual features and LSTMs for sequence modeling—a distinct influence on the proposed architecture of Smashifix. Other significant architectures include Two-Stream Networks by Simonyan and Zisserman [21] and Inflated 3D ConvNets (I3D) by Carreira and Zisserman [22]. Recent advancements have also explored the use of Slow-Fast networks, which use a dual-pathway approach to capture both spatial semantics and high-resolution motion details [23].

2.4 Lightweight Architectures and Attention Mechanisms

The computational demands of deep learning models present a barrier to real-time, on-device deployment. To address this, lightweight architectures have gained significant attention.

Ma et al. [24] proposed ShuffleNet V2, a set of practical design guidelines that introduced the channel shuffle operation. This principle forms the architectural basis for the Residual-Shuffle Network (RSN) backbone used in the Smashifix system. In the context of sports analytics, Samma and Bin Sama [25] demonstrated that an Enhanced ShuffleNet backbone could be effectively used for human action recognition. Additionally, GhostNet architectures have shown that “cheap” operations can generate more feature maps with fewer parameters, further optimizing real-time sports analysis [26].

Simultaneously, the integration of attention mechanisms into temporal models has proven highly effective. Pham et al. [27] proposed a lightweight Shuffle Graph Convolutional Network (Shuffle-GCN) that employs channel split and channel shuffle operations on skeleton data. Yuan et al. [28] further explored this by proposing a lightweight graph convolutional network with multi-attention mechanisms for action recognition.

The attention mechanism, particularly Multi-Head Self-Attention [29], allows models to dynamically weigh the importance of different time steps or spatial features in a sequence. This is supported by Yan et al. [30], who demonstrated that temporal attention significantly improves the recognition of fine-grained athletic movements. Furthermore, to ensure the interpretability of these visual features, techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) [31] have become essential.

2.5 Feedback and Biomechanical Systems

While classification is well-studied, automated corrective feedback for high-velocity sports remains an open challenge. Bhosale et al. [32] developed a yoga correction system using PoseNet and KNN, successfully classifying static asanas. However, its reliance on static pose analysis makes it unsuitable for badminton. Similarly, Harini et al. [33] used rule-based geometric heuristics for squat correction.

For more advanced analysis, Q. Wang [34] utilized a Stacked Hourglass network to refine posture detection. Crucially, Stenum and Ishøi [35] validated the use of 2D monocular video for approximating 3D kinematic parameters. Furthermore, Nakashima and Murai [36] applied deep learning to quantify baseball pitching mechanics. Liu et al. [37] proposed deep metric learning for motion comparison, a methodology mirrored in this study. In the context of skill assessment, Doughty et al. [38] pioneered the use of pairwise ranking to determine the quality of an action, while Pan et al. [39] utilized joint-relation-aware networks to evaluate student performance in physical education, providing a precedent for the KSI scoring logic.

2.6 Comparative Analysis with Existing Solutions

To contextualize the proposed Smashifix system, a comparison was conducted against existing methods. Existing systems generally excel at either classification or static feedback, but rarely both in a dynamic context. The proposed system fills this void by offering an integrated, dual-stream solution covering both classification and biomechanical correction, as summarized in Table 1 in Section 3.

3 Proposed System

3.1 System Proposal

The Smashifix system is proposed as a comprehensive coaching assistant designed to bridge the gap between amateur players and professional analysis. The system accepts monocular video footage from a standard smartphone camera. It automatically segments the video into distinct stroke instances, classifies the stroke type, and performs a forensic biomechanical analysis. The core proposition is the generation of a Kinematic Similarity Index (KSI), which mathematically quantifies how closely a user’s movement mirrors that of an elite professional, providing specific, drill-based recommendations for improvement.

3.2 System Architecture

The solution is architected around a Late-Fusion Dual-Stream Hybrid Model designed to deliver granular, high-fidelity biomechanical feedback. As illustrated in Fig. 1, the architecture processes two distinct modalities simultaneously before making a final classification:

1. **Visual Stream:** This stream extracts contextual features directly from cropped video frames. It leverages a custom Residual-Shuffle Network (RSN) as a lightweight backbone to generate independent 64-dimensional feature vectors per frame.
2. **Pose Stream:** This stream exclusively processes 33 three-dimensional skeletal landmarks extracted by MediaPipe, yielding a 99-dimensional pose vector per frame, to capture pure biomechanical movement.
3. **Intelligent Feedback Engine and Natural Language Coach:** The solution moves beyond simple error detection by introducing an enhanced Kinematic Similarity Index (KSI). This engine treats sports motion as a temporal sequence, comparing user movements against a library of expert biomechanical templates using 32 hierarchical features. It accounts for variations in speed and body type through DTW alignment with Sakoe-Chiba band constraints. A Natural Language Coach maps numerical deviations to specific biomechanical causes and generates progressive, skill-appropriate textual drills.

Fig. 1 illustrates the overview of the proposed Smashifix dual-stream architecture, detailing the parallel processing of visual and pose data pathways prior to late fusion, and the integration of the KSI feedback engine.

Table 1 Comparison of Proposed System with Existing Literature

Feature	Tukino et al. [17]	Bhosale et al. [32]	Proposed (Smashifix)
Domain	Badminton	Yoga	Badminton
Core Tech.	CNN + BlazePose	PoseNet + KNN	Attention-TCN + RSN
Input Data	Dynamic Video	Static Images	Dynamic Video
Objective	Classification	Posture Correction	Classification + Correction
Feedback	None	Visual Overlay	Biomechanical (KSI)
Latency	Moderate	High	Low
Analysis	Basic	Static Geometry	Forensic Biomechanics

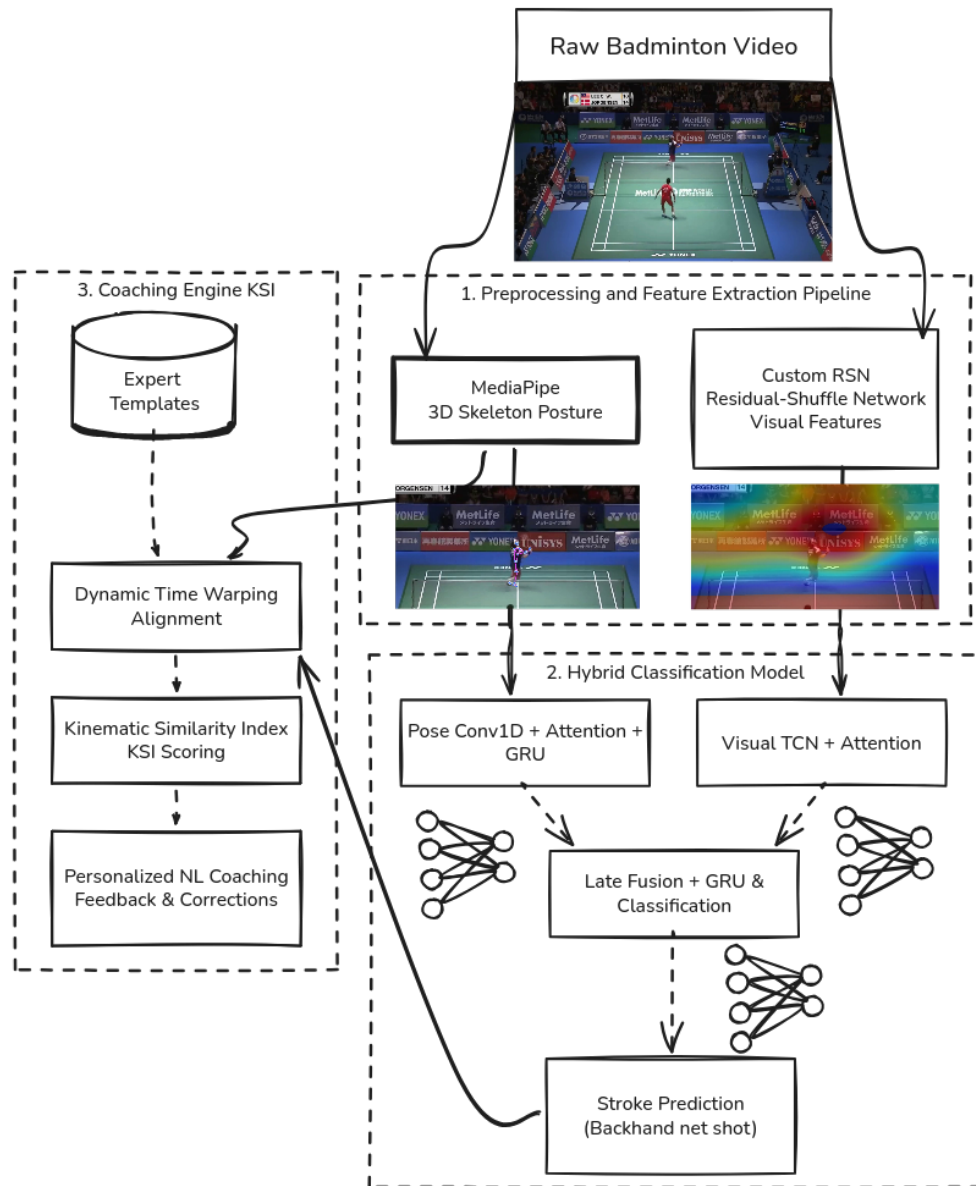


Fig. 1 Smashifix Dual-Stream Architecture

4 Methodology

This section details the technical implementation, mathematical formulations, and algorithmic logic that power the Smashifix solution.

4.1 Data Preprocessing

4.1.1 Video Segmentation

To reduce noise from pre-shot and post-shot idling, the system processes fixed-duration segments rather than entire clips. Let a video have FPS f and total frames N . Given a target segment length s (typically 1.75s–2.0s), the segment length in frames is $M = \lfloor s \cdot f \rfloor$.

For standard strokes such as Smashes and Drives, *Tail-Window Extraction* is employed to capture the follow-through, as given in Eq. (1):

$$t_0 = \max(0, N - M), \quad t \in [t_0, t_0 + M) \quad (1)$$

where:

- t_0 : The start frame index of the segment.
- N : The total number of frames in the raw video clip.
- M : The calculated window size in frames representing the action duration.

For specific defensive shots like Lifts, *Middle-Window Extraction* is utilized to center the contact point, as expressed in Eq. (2):

$$t_0 = \max(0, \lfloor \frac{N}{2} \rfloor - \lfloor \frac{M}{2} \rfloor) \quad (2)$$

4.1.2 Configuration-Driven Spatial Cropping

To stabilize pose detection and enforce consistency across varying camera placements, a deterministic Region of Interest (ROI) cropping strategy is applied. Unlike dynamic bounding boxes that rely on joint extrema and suffer from temporal jitter, this approach utilizes static, fractional bounding boxes defined systematically per shot type and video sequence.

For each video, a crop configuration dictionary specifies fractional inset values C_{top} , C_{bottom} , C_{left} , and C_{right} relative to the original frame dimensions W and H . Given a frame at index t , the vertical bounds of the extracted ROI I_t^{cropped} are defined by Eq. (3):

$$y_1 = \lfloor H \cdot C_{\text{top}} \rfloor, \quad y_2 = H - \lfloor H \cdot C_{\text{bottom}} \rfloor \quad (3)$$

and the horizontal bounds are given by Eq. (4):

$$x_1 = \lfloor W \cdot C_{\text{left}} \rfloor, \quad x_2 = W - \lfloor W \cdot C_{\text{right}} \rfloor \quad (4)$$

This configuration is resolved hierarchically:

1. **Global Baseline:** A default spatial window is established, preserving the entire frame horizontally while cropping extraneous vertical space.
2. **Shot-Specific Overrides:** Distinct stroke categories require different spatial regions of focus. For example, an overhead clear requires upper-frame attention whereas a net shot requires full-body visibility. The system applies tailored crop subsets based on recognized stroke patterns.
3. **Sequence Exceptions:** For videos with anomalous camera distances or angles, granular video-index overrides are applied to ensure consistent player scaling across the dataset.

Furthermore, the preprocessing algorithm actively detects standard naming conventions indicating pre-cropped sequences, bypassing spatial reduction to avoid destructive secondary cropping. This mathematically deterministic spatial window guarantees that downstream feature extraction networks receive temporally coherent, jitter-free visual arrays.

Fig. 2 visualizes the hybrid feature extraction pipeline for representative samples across all six shot classes. Each row shows, from left to right, the original video frame with ROI cropping applied, the 33-landmark MediaPipe Pose skeleton forming the 99-dimensional pose stream, and the RSN Grad-CAM activation map forming the 64-dimensional visual stream. The concatenation of these two complementary representations yields the 163-dimensional hybrid feature vector x_t .

4.1.3 Geometric Pose Normalization

The raw 33 three-dimensional landmark coordinates extracted by MediaPipe form a 99-dimensional vector per frame. To ensure the model learns posture geometry rather than camera position or player distance from the camera, a three-step normalization is applied to this landmark matrix $P \in \mathbb{R}^{33 \times 3}$, visualized in the center column of Fig. 2:

1. **Centering:** The skeleton is translated so the hip center c becomes the origin, as defined in Eq. (5):

$$c = \frac{P_{LH} + P_{RH}}{2}, \quad \tilde{P} = P - c \quad (5)$$

where c is the midpoint between the Left Hip (P_{LH}) and Right Hip (P_{RH}), and $\tilde{P}$ represents the translated coordinates.

2. **Alignment:** A spine vector s is defined from the hip center to the shoulder midpoint. An orthonormal basis $R = [\hat{x}; \hat{y}; \hat{z}]$ is constructed such that the spine aligns with the vertical Y-axis. The pose is rotated via $P' = \tilde{P}R^T$.
3. **Scaling:** The pose is normalized by the spine length $l = \|s\|$ to achieve scale invariance, as given in Eq. (6):

$$\bar{P} = \frac{P'}{l} \quad (6)$$

where $\bar{P}$ is the final normalized pose vector and l is the Euclidean length of the user's spine in the frame.

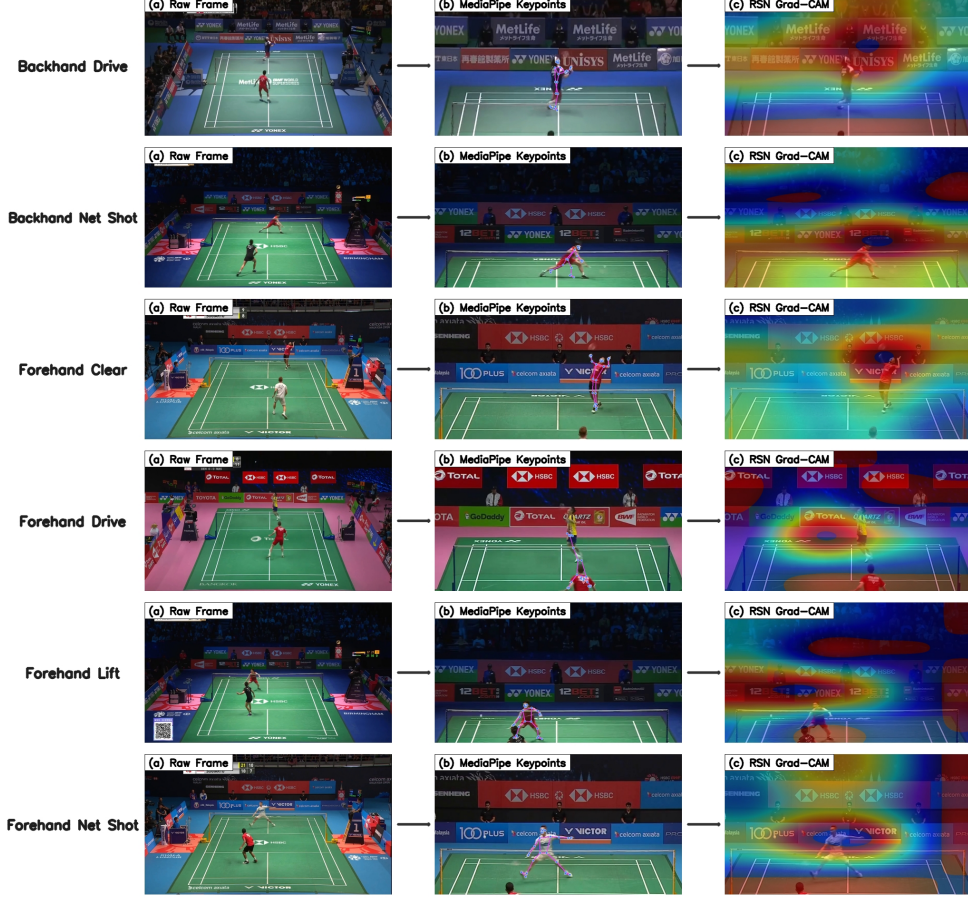


Fig. 2 Hybrid feature extraction progression for one exemplar from each of the six shot classes. Each row presents, from left to right: (a) ROI-cropped raw frame, (b) MediaPipe 33-landmark skeletal representation forming the 99-dimensional pose stream, and (c) RSN Grad-CAM activation map forming the 64-dimensional visual stream. The progression highlights how the fused representation captures both kinematic structure and discriminative visual attention before forming the 163-dimensional hybrid vector x_t .

4.2 Hybrid Feature Engineering

4.2.1 Residual-Shuffle Network (RSN) Backbone

For the Hybrid pipeline, a compact visual embedding is extracted to capture context (e.g., racket position, grip nuances). Unlike conventional approaches that use pre-trained ImageNet models such as MobileNetV2, the Smashifix system employs a custom Residual-Shuffle Network (RSN) inspired by the ShuffleNet V2 [24] design principles. The RSN is a lightweight, efficient backbone that combines residual connections with channel shuffling for effective cross-group information mixing.

The RSN architecture consists of:

- **Stem:** A 3×3 convolution with stride 2 followed by batch normalization, ReLU activation, and max pooling for initial feature extraction and spatial downsampling.
- **Shuffle Units (4 Stages):** Each stage comprises multiple Shuffle Units. For stride-1 units, the input channels are split in half: one half passes through an identity shortcut while the other is processed by a bottleneck consisting of 1×1 Conv $\rightarrow$ Depthwise 3×3 Conv $\rightarrow 1 \times 1$ Conv, each followed by batch normalization and ReLU. For stride-2 units, both branches downsample, effectively doubling the channel count. After concatenation, a channel shuffle operation redistributes information across groups.
- **Global Average Pooling + Projection:** The backbone output is reduced to a global feature vector, projected to d dimensions via a dense layer with ReLU activation, and L2-normalized.

The channel shuffle operation is described as a reshape-transpose-flatten sequence on the channel dimension, as expressed in Eq. (7):

$$\text{Shuffle}(x, g) = \text{Flatten}(\text{Transpose}(\text{Reshape}(x, [B, H, W, g, C/g]), [0, 1, 2, 4, 3])) \quad (7)$$

where g is the number of groups (set to 4) and C is the total channel count.

The complete stage-wise configuration of the RSN backbone is presented in Table 2. Channels double and spatial resolution halves at each stage, yielding a compact 512-dimensional global descriptor after Global Average Pooling.

Two-Stage RSN Training Procedure.

The RSN is trained in two phases to produce the compact 64-dimensional visual embedding that constitutes 64 of the 163 dimensions in the hybrid feature vector x_t .

Stage 1 (Supervised Pre-training). ROI-cropped frames are extracted offline from training videos and resized to $224 \times 224 \times 3$. Pixel values are rescaled to $[0, 1]$. A classification head—Dropout(0.3) $\rightarrow$ Dense(256, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(6, Softmax)—is appended to the backbone, and the entire network is trained end-to-end for six-class stroke classification. Training uses the Adam optimizer ($\eta = 10^{-3}$), sparse categorical cross-entropy loss, and three callbacks: (i) model checkpointing saving the best validation-accuracy weights, (ii) early stopping with patience 10, and (iii) ReduceLROnPlateau (factor 0.5, patience 3, minimum 10^{-6}). The data pipeline leverages cached, shuffled, and prefetched `tf.data` batches for GPU throughput.

Table 2 RSN (Residual-Shuffle Network) Stage Configuration. Each downsampling Shuffle Unit uses stride 2; subsequent units use stride 1. Channels double at each stage boundary.

Stage	Output Size	Shuffle Units	Channels
Stem (Conv 3×3, s2 + BN + ReLU + MaxPool s2)	56 × 56	–	64
Stage 1	28 × 28	2	64
Stage 2	14 × 14	3	128
Stage 3	7 × 7	3	256
Stage 4	4 × 4	2	512
Global Average Pooling	1 × 1	–	512
Projection (Dense + ReLU + L2-Norm)	–	–	$d=64$

Stage 2 (Frozen Feature Extraction). After convergence, the classification head is discarded and the backbone weights are frozen ($\theta_{\text{RSN}} \leftarrow \text{const}$). A lightweight projection head Dense($d=64$, ReLU) followed by L2-normalization replaces the classifier. For each incoming frame I_t , the frozen extractor produces a 64-dimensional unit-norm vector c_t capturing racket orientation, grip angle, and shuttle-contact vicinity without overfitting to training identities. This design mirrors transfer-learning practice while remaining computationally inexpensive (~ 0.4 GFLOPs per frame). The two-stage procedure is critical: Stage 1 forces the backbone to learn stroke-discriminative visual features, while Stage 2 produces a compact, generalizable embedding suitable for downstream temporal fusion.

The ROI-cropped frame I_t is resized to 224×224 and passed through the pre-trained RSN to produce the visual feature vector, as given in Eq. (8):

$$h_t = \text{RSN}(I_t) \in \mathbb{R}^d \quad (8)$$

where h_t is the feature map extracted from input image I_t . To interpret which spatial regions drive RSN decisions, Grad-CAM [31] is computed from the final convolutional feature tensor A^k and a target score y^c . The importance weight of each channel is computed as in Eq. (9):

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (9)$$

and the activation map is computed as in Eq. (10):

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (10)$$

where α_k^c is the importance weight of channel k and Z is the number of spatial locations. The map is then bilinearly upsampled to the input resolution and alpha-blended with the ROI frame for visualization. In the rightmost column of Fig. 2, high-intensity

regions consistently concentrate around the racket head, wrist-elbow chain, and shuttle contact vicinity, indicating that the visual stream captures stroke-discriminative biomechanics.

This embedding is projected to 64 dimensions and L2-normalized, as expressed in Eq. (11):

$$u_t = \text{ReLU}(Wh_t + b), \quad c_t = \frac{u_t}{\|u_t\|_2} \quad (11)$$

where W and b are learnable weights and biases, and c_t is the normalized context vector.

4.2.2 Feature Fusion

The normalized pose vector and the visual embedding are concatenated to form the final hybrid feature vector x_t , as defined in Eq. (12):

$$x_t = [\text{vec}(\bar{P}_t) \parallel c_t] \in \mathbb{R}^{163} \quad (12)$$

Fig. 2 summarizes the complete dual-stream extraction pipeline, from the raw cropped frame through to the pose and visual embeddings.

4.2.3 Classification Architecture

The core classification engine is a dual-stream hybrid architecture combining an Attention-enhanced Temporal Convolutional Network with a late fusion strategy:

- **Visual Branch (Attention-TCN):** The visual feature sequence undergoes an initial 1×1 projection to F filters, followed by a deep TCN consisting of 4 dilated causal Conv1D layers with dilations $d \in \{1, 2, 4, 8\}$. Each TCN layer includes batch normalization, ReLU activation, spatial dropout, and a residual connection. The exponentially increasing dilation yields a receptive field of 31 frames. The output is then processed by a Multi-Head Self-Attention layer (h heads, key dimension k) and a layer-normalized GRU block (G units).
- **Pose Branch:** The normalized pose sequence is independently processed by a Conv1D layer (F filters, kernel size 3, causal padding), followed by an identical Multi-Head Self-Attention block and a GRU layer (G units) with batch normalization.
- **Late Fusion:** The sequence summaries of both branches are concatenated to form a 163-dimensional hybrid feature vector. This vector is passed through two dense layers (F_{fuse} and $F_{\text{fuse}}/2$ units) with dropout and L2 regularization, terminating in a softmax classifier for final stroke prediction.

The dilated causal convolution for layer l with dilation rate d_l is expressed in Eq. (13):

$$\text{DilConv}(x_t, d_l) = \sum_{k=0}^{K-1} w_k \cdot x_{t-d_l \cdot k} \quad (13)$$

where K is the kernel size (set to 3), w_k are the learned filter weights, and the causal constraint ensures $t - d_l \cdot k \geq 0$.

The Multi-Head Self-Attention mechanism is formalized in Eq. (14):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (14)$$

where Q , K , and V are the query, key, and value projections of the input sequence, and d_k is the key dimension. The multi-head variant concatenates h independent attention heads and projects the result. A residual connection preserves gradient flow, as shown in Eq. (15):

$$y = \text{LayerNorm}(x + \text{MultiHead}(x, x, x)) \quad (15)$$

The hyperparameters of the Attention-TCN were optimized using Optuna [40] with a Bayesian search strategy. The tuned values are: $F = 80$ filters, $G = 80$ GRU units, $h = 8$ attention heads, $k = 32$ key dimension, $F_{\text{fuse}} = 80$ fusion units, L2 regularization $\lambda = 7.6 \times 10^{-5}$, dropout rate $p = 0.22$, learning rate $\eta = 5.6 \times 10^{-4}$, and label smoothing $\epsilon = 0.1$.

4.2.4 Contact-Aware Prediction Algorithm

To improve accuracy, the specific temporal window containing the stroke contact point is identified. A composite arm velocity v_t is calculated based on the right shoulder, elbow, and wrist. The window with the maximum acceleration magnitude A is selected for the final prediction, as given in Eq. (16):

$$A = \|\Delta v\|_2, \quad \text{Class} = \arg \max_c p(c|X_{\text{peak_acc}}) \quad (16)$$

where A is the magnitude of acceleration, and Δv represents the change in velocity vectors between consecutive frames.

4.3 Biomechanical Analysis (KSI Algorithm)

The core of the feedback mechanism is the enhanced Kinematic Similarity Index (KSI), which compares the user’s motion sequence U against a DTW-aligned expert template E . KSI extends the original algorithm with 32 hierarchical biomechanical features, contact-centered windowing, confidence filtering, and jerk analysis.

The complete KSI computation is formalized in Algorithm 1.

4.3.1 Enhanced Feature Extraction (32D)

KSI extracts 32 hierarchical biomechanical features per frame, organized into six categories:

- **Joint Angles (12):** Elbow flexion, shoulder elevation, shoulder abduction, knee flexion, hip flexion, and ankle dorsiflexion for both left and right sides.
- **Limb Ratios (6):** Full arm extension, full leg extension, and forearm-to-upper-arm ratio, normalized by torso length for scale invariance.
- **Spinal Alignment (4):** Forward lean (sagittal), lateral lean (frontal), spine twist (transverse), and normalized spine length.

- **Rotational Dynamics (4):** Hip-shoulder separation angle (critical for power generation), hip rotation, shoulder rotation, and trunk rotation velocity.
- **Wrist Dynamics (3):** Wrist height, horizontal reach, and radial deviation—all normalized relative to the torso.
- **Center of Mass (3):** Approximate CoM displacement in x , y , and z axes, computed as a weighted average of hip (50%), shoulder (30%), and knee (20%) positions.

++++

4.3.2 Confidence Filtering

Before analysis, unreliable frames are filtered based on two criteria:

1. **Anatomical Plausibility:** Limb length ratios are checked against expected anthropometric ranges. Frames with anatomically implausible configurations are rejected.
2. **Temporal Consistency:** Frames exhibiting sudden landmark displacements exceeding 0.3 (in normalized coordinates) between consecutive frames are flagged as tracking artifacts and excluded.

4.3.3 Contact-Centered Windowing

To focus computational resources on the most biomechanically relevant portion of the sequence, a contact-centered window is applied. The contact frame t_c is identified from the shot phase segmentation, and a window of $\pm W$ frames is extracted around it, as defined in Eq. (17):

$$\text{Window} = [t_c - W_{\text{pre}}, t_c + W_{\text{post}}] \quad (17)$$

where W_{pre} and W_{post} are the pre-contact and post-contact window sizes (set to 18 frames each, covering approximately 600ms at 30 FPS).

4.3.4 Dynamic Time Warping (DTW) with Sakoe-Chiba Band

To account for temporal variations in execution speed, DTW aligns the sequences by minimizing the cumulative Euclidean distance. The recurrence relation for the cost matrix D is given in Eq. (18):

$$D_{i,j} = \|F_i^E - F_j^U\|_2 + \min\{D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}\} \quad (18)$$

where:

- $D_{i,j}$: The cumulative cost to align Expert frame i with User frame j .
- F_i^E, F_j^U : The biomechanical feature vectors for the Expert and User respectively.
- $\|\cdot\|_2$: The Euclidean distance metric.

To improve computational efficiency and prevent pathological alignments, a Sakoe-Chiba band constraint [41] is applied, restricting the warping path to a band of width

Algorithm 1 Kinematic Similarity Index (KSI)

Require: User landmarks $\mathbf{U} \in \mathbb{R}^{T_U \times 33 \times 3}$, Expert template $\mathbf{E} \in \mathbb{R}^{T_E \times 33 \times 3}$, Weights $\mathbf{w} = (w_p, w_v, w_a)$, Confidence threshold τ , Band ratio β

Ensure: KSI score $S \in [0, 1]$, Per-joint error map $\mathcal{E}$, Feedback vector $\mathbf{F}$

```
1: — Stage 1: Confidence Filtering —
2: for each frame  $t$  in  $\mathbf{U}$  and  $\mathbf{E}$  do
3:   Reject if mean landmark visibility  $< \tau$ 
4:   Reject if any limb-length ratio falls outside anthropometric bounds
5:   Reject if  $\max_j \|\mathbf{p}_j^t - \mathbf{p}_j^{t-1}\|_2 > 0.3$  ▷ temporal artifact
6: end for
7:  $\hat{\mathbf{U}} \leftarrow$  valid user frames;  $\hat{\mathbf{E}} \leftarrow$  valid expert frames

8: — Stage 2: Shot-Phase Segmentation & Contact Windowing —
9:  $v_t \leftarrow \|\Delta \hat{\mathbf{E}}_{\text{R\_wrist}}\|$  ▷ racket-arm wrist velocity
10:  $t_c \leftarrow \arg \max_t v_t$  ▷ contact frame
11:  $\tilde{\mathbf{E}} \leftarrow \hat{\mathbf{E}}[t_c - W_{\text{pre}} : t_c + W_{\text{post}}]$ 
12:  $\hat{\mathbf{U}} \leftarrow$  similarly windowed user sequence
13: Segment into phases: PREPARATION, LOADING, ACCELERATION, CONTACT, FOLLOW-THROUGH

14: — Stage 3: Hierarchical Feature Extraction (32-D) —
15: for each frame  $t$  in  $\hat{\mathbf{U}}$  and  $\tilde{\mathbf{E}}$  do
16:    $\mathbf{f}_t \leftarrow \text{EXTRACT}_{32}(\text{angles, ratios, spine, rotation, wrist, CoM})$ 
17: end for
18: Compute trunk rotation velocity:  $\dot{\phi}_t \leftarrow \nabla_t(\text{hip-shoulder separation})$ 

19: — Stage 4: DTW Alignment with Sakoe-Chiba Band —
20:  $r \leftarrow \lfloor \beta \cdot \max(T_E, T_U) \rfloor$  ▷ band width
21: Fill cost matrix  $D_{i,j}$  subject to  $|i - j| \leq r$ 
22: Backtrack optimal path  $\mathcal{W}^*$ ; produce aligned pair  $(\tilde{\mathbf{E}}, \tilde{\mathbf{U}})$ 

23: — Stage 5: Phase-Aware Attention Weights —
24: for each aligned frame  $k$  do
25:    $\alpha_k \leftarrow w_{\text{phase}} \cdot (0.5 + 0.5 \exp(-(k - \mu_{\text{phase}})^2 / 2\sigma_{\text{phase}}^2))$ 
26: end for

27: — Stage 6: Multi-Component Scoring —
28:  $S_{\text{pose}} \leftarrow \sum_k \alpha_k \cos(\tilde{\mathbf{E}}_k, \tilde{\mathbf{U}}_k) / \sum_k \alpha_k$ 
29:  $S_{\text{vel}} \leftarrow \sum_k \alpha_k \cdot \cos_{\text{vel},k} \cdot \exp(-0.1 \|V_k^E - V_k^U\|^2) / \sum_k \alpha_k$ 
30:  $S_{\text{acc}} \leftarrow$  analogous formulation on second derivatives
31:  $S \leftarrow w_p \cdot S_{\text{pose}} + w_v \cdot S_{\text{vel}} + w_a \cdot S_{\text{acc}}$ 

32: — Stage 7: Per-Joint Error Analysis & Feedback —
33: for each feature  $i \in \{1, \dots, 32\}$  do
34:    $\mathcal{E}_i \leftarrow (\bar{e}_i, e_i^{\text{max}}, \sigma_i, t_i^{\text{crit}}, \text{CI}_{95\%})$  via adaptive bootstrap
35:   if  $\bar{e}_i > \theta_{\text{severity}}$  then
36:     Append biomechanical correction  $\delta_i$  to  $\mathbf{F}$  via causal-chain lookup
37:   end if
38: end for
39: return  $S, \mathcal{E}, \mathbf{F}$ 
```

r around the diagonal, as expressed in Eq. (19):

$$|i - j| \leq r, \quad r = \lfloor \beta \cdot \max(N_E, N_U) \rfloor \quad (19)$$

where β is the band ratio (set to 0.2) and N_E, N_U are the lengths of the expert and user sequences respectively.

4.3.5 Phase-Aware Attention

The system segments the aligned shot into five biomechanical phases: *Preparation*, *Loading*, *Acceleration*, *Contact*, and *Follow-through*. An attention weight α_k is assigned to each frame using a Gaussian envelope centered on the phase, as expressed in Eq. (20):

$$\alpha_k = w_{\text{phase}} \cdot \left(0.5 + 0.5 \exp \left(-0.5 \left(\frac{k - \text{center}}{\text{width}} \right)^2 \right) \right) \quad (20)$$

where:

- α_k : The importance weight assigned to frame k .
- w_{phase} : The predefined weight of the current phase, with Contact phase $w = 3.0$, Acceleration phase $w = 2.0$, and Preparation phase $w = 0.8$.
- center: The frame index representing the midpoint of the phase.
- width: A parameter controlling the spread of the Gaussian curve.

4.3.6 Scoring Components

The final KSI score is an aggregate of four differential kinematic similarities.

Pose Similarity (S_{pose}): Computed as the attention-weighted cosine similarity of the feature vectors, as given in Eq. (21):

$$S_{\text{pose}} = \frac{\sum_k \alpha_k \cdot \frac{F_k^E \cdot F_k^U}{\|F_k^E\| \cdot \|F_k^U\|}}{\sum_k \alpha_k} \quad (21)$$

Velocity Similarity (S_{vel}): Incorporates both directional alignment and magnitude matching, as defined in Eq. (22):

$$S_{\text{vel}} = \frac{\sum_k \alpha_k \cdot \cos_{\text{vel},k} \cdot \exp(-0.1 \|V_k^E - V_k^U\|^2)}{\sum_k \alpha_k} \quad (22)$$

where $\cos_{\text{vel},k}$ is the cosine similarity between the User’s and Expert’s velocity directions at frame k , and V^E, V^U are the respective velocity magnitudes.

Acceleration Similarity (S_{acc}) and **Jerk Similarity (S_{jerk})** follow analogous formulations using the second and third temporal derivatives of the feature sequences, respectively.

The final weighted KSI score is computed as given in Eq. (23):

$$\text{KSI}_{\text{total}} = w_p \cdot S_{\text{pose}} + w_v \cdot S_{\text{vel}} + w_a \cdot S_{\text{acc}} \quad (23)$$

where $w_p = 0.4$, $w_v = 0.4$, and $w_a = 0.2$ are the component weights. As illustrated in Fig. 5, the detailed model architecture and end-to-end prediction pipeline show the flow from video input through pose estimation, feature extraction, dual-pipeline classification, and KSI-based biomechanical feedback generation.

4.3.7 Natural Language Coaching Generation

While the KSI provides a rigorous mathematical foundation for identifying errors, numerical indices alone lack the actionable context required by amateur players. To bridge this gap, Smashifix integrates a Natural Language Coach that transforms raw KSI outputs into personalized, pedagogically structured coaching reports. The module is composed of three principal components: a Biomechanical Knowledge Base, a multi-stage feedback generation pipeline, and a skill-adaptive language engine.

Biomechanical Knowledge Base.

The foundation of the coach is an expert-authored knowledge base encoding domain-specific biomechanical causality chains. For each of the 32 features monitored by the KSI (Algorithm 1), the knowledge base stores: (i) a set of observable *symptoms* (e.g., “elbow too straight”, “arm dropping”), (ii) a causal chain mapping each symptom to underlying biomechanical causes—for instance, the racket arm elbow dropping may stem from rotator cuff fatigue, habitual bad form, or an excessively tight grip—(iii) the governing biomechanical principle explaining the kinetic chain role of that joint (e.g., “the elbow acts as a velocity multiplier; optimal angle 90–120° allows maximum angular velocity transfer from shoulder to wrist”), and (iv) shot-specific priority rankings indicating which features are most critical for each of the six stroke types.

Error Classification and Causal Reasoning.

The KSI per-joint error map $\mathcal{E}$ is first classified into four severity tiers using configurable thresholds on the effective error magnitude:

- **Critical** ($\geq 25^\circ$): Fundamental biomechanical fault requiring immediate correction.
- **Major** ($\geq 15^\circ$): Significant deviation impacting power or consistency.
- **Minor** ($\geq 8^\circ$): Detectable deviation with marginal performance impact.
- **Negligible** ($< 8^\circ$): Within acceptable biomechanical tolerance.

Shot-specific priority features receive a $1.5\times$ severity boost (top two priorities) or $1.2\times$ (secondary priorities), so that wrist contact height and hip-shoulder separation are weighted more heavily for a Forehand Clear than for a Net Shot. For the top three errors (two Critical, one Major), the coach traverses the causal chain to determine the most likely root cause and selects the corresponding correction strategy.

Skill-Level Adaptive Language.

The system supports four skill levels—Beginner, Intermediate, Advanced, and Expert—and dynamically adjusts its vocabulary, explanatory depth, and prescribed drills accordingly. Table 3 illustrates this adaptation for correcting insufficient hip-shoulder separation:

Table 3 Skill-Level Adaptation: Hip-Shoulder Separation Correction

Level	Generated Correction
Beginner	“Turn your body like you’re throwing a ball sideways. Hips face forward first, then shoulders catch up.”
Intermediate	“Create ‘X-factor’ stretch: rotate hips 30° toward target while shoulders stay back, then unwind explosively.”
Advanced	“Target 15–25° separation at peak. Initiate with hip rotation from ground force; let shoulders lag 40–60 ms.”
Expert	“Optimize stretch-shortening cycle timing. Peak separation should occur 80–100 ms before contact.”

Each correction is accompanied by a structured coaching item containing: (i) the issue and severity rating, (ii) the root-cause explanation from the causal chain, (iii) a skill-appropriate fix, (iv) a specific drill with equipment requirements and duration, (v) a visual mental cue (e.g., “picture wringing a wet towel”), (vi) common mistakes to avoid, and (vii) observable success indicators. Beyond individual corrections, the coach generates phase-by-phase analysis (Preparation through Follow-Through), timing feedback based on peak-velocity offset relative to the expert template, power feedback derived from velocity ratios, a progressive four-week training plan, and a motivational closing message.

Example System Output.

Table 4 presents an abridged coaching report generated by Smashifix for a real user performing a Backhand Net Shot at the Intermediate skill level, with a KSI score of $S = 0.35$.

This structured output demonstrates how the system converts a single numerical KSI score into a rich, multi-dimensional coaching narrative that addresses *what* is wrong, *why* it occurs, and *how* to fix it—with progressive specificity adapted to the player’s current ability. The full report additionally includes a four-week training plan with week-by-week drill progressions, video timestamp annotations for key error frames, and—for Advanced/Expert users—a technical appendix listing raw component scores (S_{pose} , S_{vel} , S_{acc} , S_{jerk}), DTW distance, and 95% bootstrap confidence intervals.

Table 4 Abridged Coaching Report (Backhand Net Shot, KSI = 0.35, Intermediate)

Field	Generated Content
Rating	★ Foundation Building
Summary	“Let’s rebuild your Backhand Net Shot from the ground up. Your score of 0.35 indicates fundamental technique issues, but small changes will create big improvements!”
Priority Fix	Trunk Posture (151.8° deviation, Critical): “Maintain 5–10° forward lean at contact. Core engaged (like bracing for a punch). Excessive lean = weak shots.” Drill: Broomstick overhead shadow footwork, 5 min/session.
Timing	“Early peak: reaching maximum speed 533 ms too early. Delay your forward swing initiation.”
Power	“Excellent power generation! Matching expert racket head speed.”
Strengths	Arm extension positioning ($< 5^\circ$ error); acceleration phase score 0.85.

5 Experiments and Results

5.1 Dataset Description

The system was trained on the Badminton Stroke Video Dataset sourced from Kaggle [42], augmented with proprietary high-definition footage. The dataset comprises 1.32 GB of data capturing human subjects performing strokes from multiple angles under consistent lighting conditions. It is categorized into six distinct stroke classes: Forehand Clear, Forehand Drive, Forehand Lift, Forehand Net Shot, Backhand Drive, and Backhand Net Shot. The data was partitioned using a stratified split into a Training Set of 70%, a Validation Set of 15%, and a Test Set of 15%.

Representative frames from each stroke class are shown in Fig. 3.

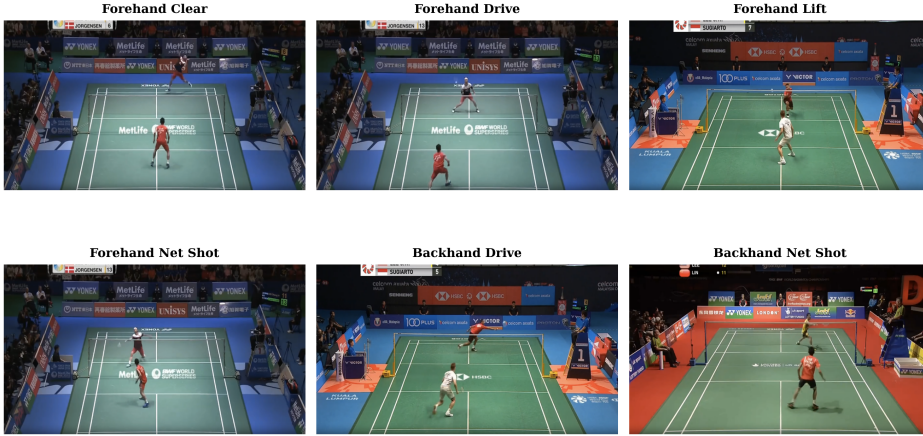


Fig. 3 Dataset Sample Frames: one representative frame from each of the six badminton stroke classes, captured from monocular video under consistent lighting conditions.

Fig. 4 presents the distribution of samples across all six stroke classes for the training, validation, and test splits, demonstrating the balanced representation achieved through stratified sampling.

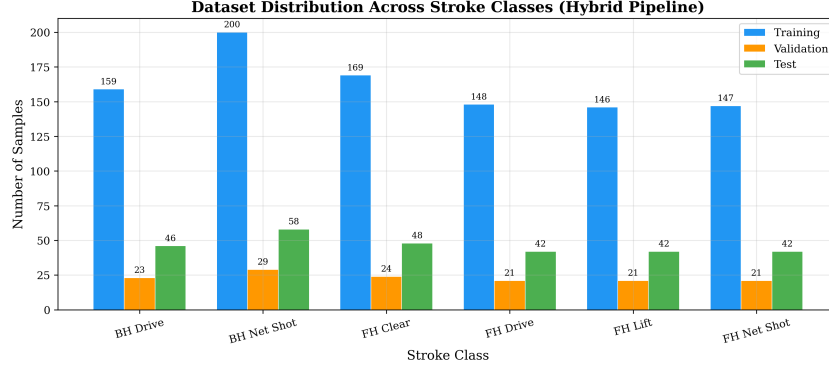


Fig. 4 Class Distribution Across Splits

Fig. 5 depicts the end-to-end prediction pipeline of the Smashifix system, showing the complete flow from video input through preprocessing, dual-stream feature extraction, classification, and KSI-driven feedback generation.

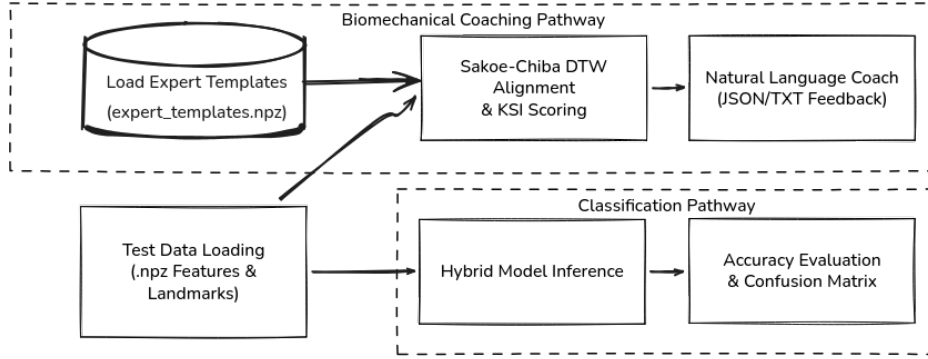


Fig. 5 End-to-End Prediction Pipeline: the complete flow from raw video input through spatial cropping, pose estimation, RSN visual feature extraction, dual-stream Attention-TCN classification, and KSI-based Natural Language feedback generation.

5.2 Evaluation Metrics

To rigorously assess the performance of the classification pipelines, the following metrics were employed:

1. **Accuracy:** The ratio of correctly predicted observations to the total observations, as defined in Eq. (24):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (24)$$

2. **Precision:** The ratio of correctly predicted positive observations to the total predicted positives, as given in Eq. (25):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (25)$$

3. **Recall:** The ratio of correctly predicted positive observations to all observations in the actual class, as given in Eq. (26):

$$\text{Recall} = \frac{TP}{TP + FN} \quad (26)$$

4. **F1-Score:** The weighted average of Precision and Recall, as computed in Eq. (27):

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (27)$$

where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

5.3 Quantitative Results

The performance of the proposed pipelines on the held-out test set (20% stratified split, $n = 278$) is summarized in Table 5. The Attention-TCN pipeline demonstrated superior performance due to the inclusion of visual context features and temporal self-attention, achieving a test accuracy of 98.56% and a macro-averaged F1-Score of 0.98, significantly outperforming both the pose-only and visual-only baselines.

Table 5 Classification Performance Results

Model Architecture	Accuracy	Precision	Recall	F1-Score
Pose-LSTM (Real-time)	0.81	0.79	0.78	0.78
Attention-TCN (Proposed)	0.9856	0.99	0.98	0.98
RSN (Visual only)	0.74	0.72	0.71	0.71

To provide granular insight into the model’s discriminative capability across individual stroke types, Table 6 presents the per-class precision, recall, and F1-score on the same test set. All six classes achieve F1-scores ≥ 0.98 , with Forehand Clear attaining

perfect scores. The slight reduction in recall for Forehand Drive (0.95) indicates occasional confusion with visually similar strokes such as Forehand Lift, which remains a known challenge in badminton action recognition.

Table 6 Attention-TCN Per-Class Classification Metrics

Shot Class	Precision	Recall	F1-Score	Support
Backhand Drive	0.96	1.00	0.98	46
Backhand Net Shot	0.98	1.00	0.99	58
Forehand Clear	1.00	1.00	1.00	48
Forehand Drive	1.00	0.95	0.98	42
Forehand Lift	1.00	0.98	0.99	42
Forehand Net Shot	0.98	0.98	0.98	42
Macro Average	0.99	0.98	0.98	278

5.4 Training Dynamics

The training dynamics of both pipelines are illustrated through their epoch-wise accuracy and loss curves. Fig. 6 shows the Attention-TCN training over 193 epochs, reaching a peak validation accuracy of 94.70% at epoch 165. The model exhibits slight overfitting in later epochs, with training accuracy exceeding validation accuracy, though the gap remains modest. Fig. 7 depicts the Pose-LSTM training over 71 epochs, converging to a best validation accuracy of 90.44% at epoch 59, demonstrating faster but lower-ceiling convergence compared to the TCN.

Fig. 6 illustrates the training and validation curves for the Attention-TCN pipeline over 193 epochs: (a) accuracy progression showing peak validation accuracy of 94.70% at epoch 165, and (b) corresponding loss curves demonstrating consistent convergence.

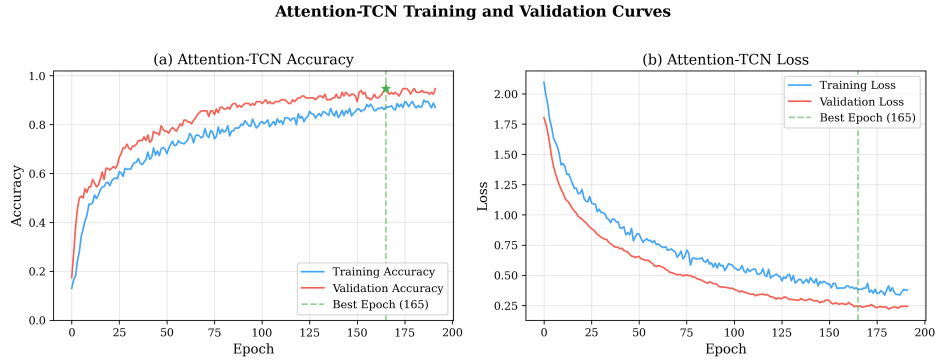


Fig. 6 Attention-TCN Training Curves

Fig. 7 presents the training and validation curves for the Pose-LSTM pipeline over 71 epochs: (a) accuracy progression showing peak validation accuracy of 90.44% at epoch 59, and (b) corresponding loss curves with characteristic LSTM oscillation during convergence.

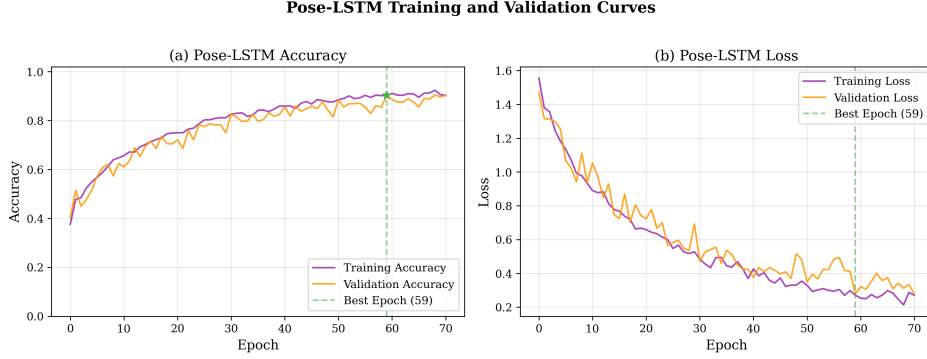


Fig. 7 Pose-LSTM Training Curves

A direct comparison of the validation accuracy trajectories of both models is presented in Fig. 8. The overlay reveals that while the Pose-LSTM initially learns faster due to its simpler architecture and lower-dimensional input, the Attention-TCN ultimately surpasses it after approximately 60 epochs, benefiting from the richer visual context and temporal attention mechanism.

As depicted in Fig. 8, an overlay comparison of validation accuracy between the Attention-TCN and Pose-LSTM pipelines illustrates the TCN's superior long-term convergence despite the LSTM's faster initial learning rate.

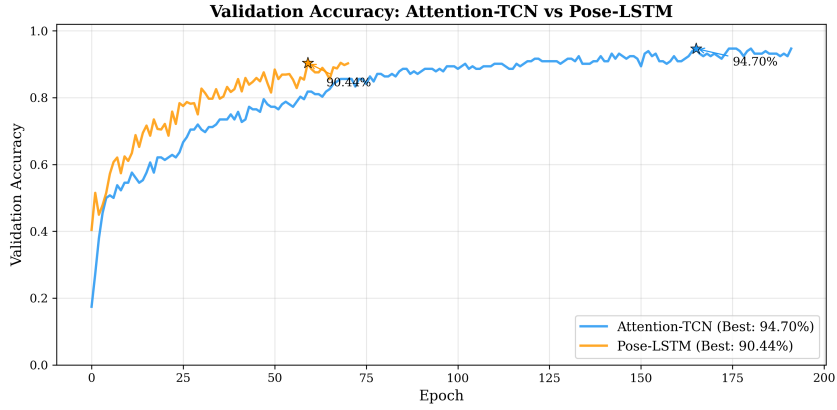


Fig. 8 Validation Accuracy Overlay Comparison

5.5 Hyperparameter Optimization

The Attention-TCN architecture was optimized using Optuna with Bayesian Tree-structured Parzen Estimator (TPE) sampling over 46 trials. The search space included learning rate, number of TCN filters, attention heads, key dimension, GRU units, dropout rates, and batch size. Fig. 9 presents the validation accuracy achieved by each trial, with a running-best line showing the optimization trajectory. The best configuration achieved 98.92% validation accuracy at trial 13, after which the marginal returns from additional trials diminished.

Fig. 9 shows the Optuna hyperparameter optimization results for the Attention-TCN over 46 trials using TPE sampling, with bars colour-coded by validation accuracy and a running-best line showing convergence to 98.92%.

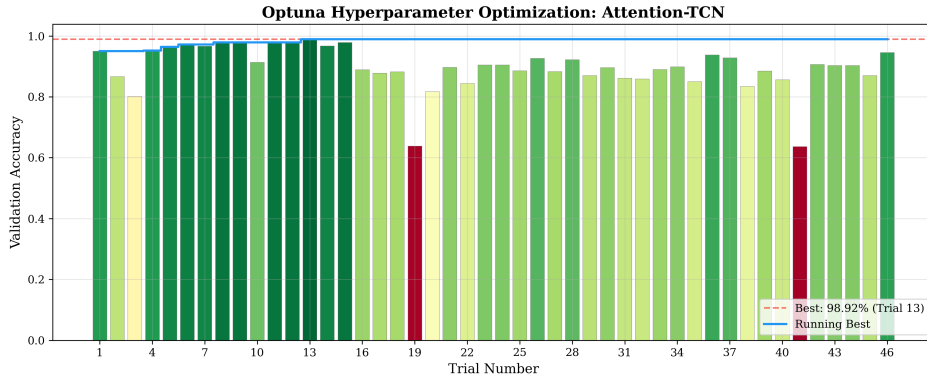


Fig. 9 Optuna Hyperparameter Optimization Results

5.6 Comparative Analysis

To validate the effectiveness of the Smashifix system, the results were compared against the methods discussed in the Literature Survey. Table 7 highlights that the Smashifix system now surpasses existing specialized classification models in accuracy while simultaneously providing dynamic biomechanical feedback granularity that other systems lack.

Table 7 Comparative Analysis with State-of-the-Art Methods

Method	Accuracy	Feedback Type	Scope
Smashifix (Proposed)	98.6%	Dynamic Biomechanical	Complex Sequence
Asriani et al. [18]	98.6%	None	Single Stroke
Tukino et al. [17]	89.2%	None	Single Stroke
Bhosale et al. [32]	82.0%	Static Overlay	Static Pose

To further validate the approach, Table 8 compares the proposed system against methods from other sporting domains, demonstrating the generalizability of the dual-pipeline and biomechanical feedback approach.

Table 8 Cross-Sport Comparative Analysis

Method	Sport	Accuracy	Feedback	Input
Siddiqui et al. [19]	Cricket	99.7%	None	Skeleton
Harini et al. [33]	Squat Exercise	91.0%	Rule-based	Skeleton
Smashifix (Proposed)	Badminton	98.6%	KSI + DTW	Skeleton + Visual

5.7 System Output Demonstration

To provide concrete visual evidence of the deployed Smashifix application, this subsection presents representative screenshots captured from the web-based interface during a live analysis session of a Forehand Clear stroke.

Fig. 10 displays the Analysis Results panel generated after forensic-mode processing. The system correctly predicts the stroke as a *Forehand Clear* with 77.8% confidence, identifies the contact frame (Frame 13), and produces a structured Coaching Feedback report. The KSI score of $S = 0.30$ places the user in the “Foundation Building” tier, triggering a set of prioritized biomechanical corrections—in this case, a critical 170.0° deviation in Spine Lateral Lean. Each detected issue is accompanied by a specific fix recommendation and a targeted drill, demonstrating the Natural Language Coach’s ability to translate numerical KSI outputs into actionable, skill-appropriate coaching advice.

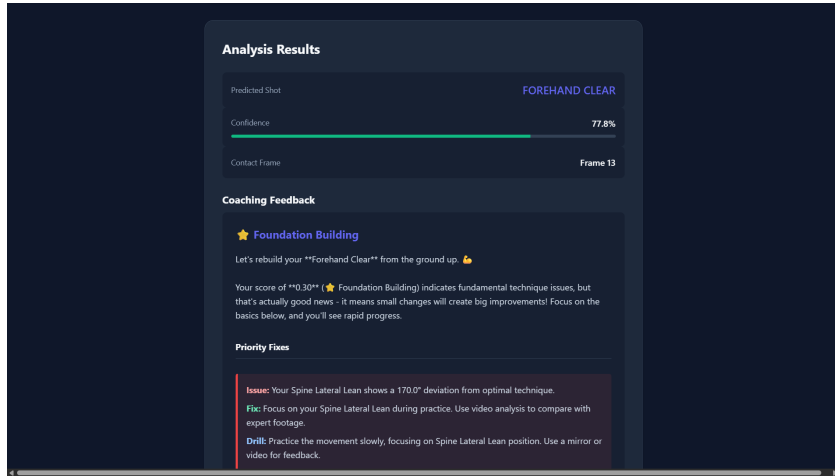


Fig. 10 Smashifix Analysis Results panel showing stroke classification (Forehand Clear, 77.8% confidence), contact frame identification, and KSI-driven coaching feedback with prioritized biomechanical corrections and drill recommendations.

Fig. 11 presents the Error Visualization module, which pinpoints specific biomechanical faults detected during the stroke execution. The interface lists time-stamped error entries (at 00:02.86s, Frame 86) identifying five concurrent deviations: spine lateral lean, spine forward lean, shoulder rotation angle, hip rotation angle, and right knee angle. This granular, frame-level error localization enables players to scrub directly to the problematic moment in the video playback, facilitating targeted review of the exact instant where technique breakdown occurs.



Fig. 11 Error Visualization panel displaying time-stamped, frame-level biomechanical error detection. Five concurrent deviations are identified at Frame 86, enabling precise temporal localization of technique faults for targeted review.

Fig. 12 illustrates the Correct Execution (Slow Motion) module, which provides a side-by-side expert reference for the classified stroke. After analysis, the system retrieves and presents a slow-motion video of an elite player performing the same stroke type (in this instance, a Forehand Clear), enabling direct visual comparison between the user’s execution and the expert template. This pedagogical feature complements the quantitative KSI feedback by offering an intuitive visual benchmark, allowing players to observe proper biomechanics—including racket arm trajectory, body rotation timing, and follow-through form—at reduced playback speed.

Collectively, these three output components demonstrate the end-to-end capability of the Smashifix system to transform a single monocular video input into a comprehensive, multi-modal coaching report. Fig. 10 shows the quantitative analysis with coaching feedback, Fig. 11 illustrates the temporal error localization, and Fig. 12 provides the expert visual reference. Together, these components address *what* was performed, *where* errors occurred, *how* to correct them, and *what* correct execution looks like.

6 Conclusion

This paper presented Smashifix, a comprehensive automated coaching system for badminton that bridges the gap between amateur practice and professional-grade biomechanical analysis. The proposed Late-Fusion Dual-Stream Hybrid architecture fuses visual features from a custom Residual-Shuffle Network backbone with 99-dimensional skeletal pose features via an Attention-enhanced Temporal Convolutional Network, delivering high-fidelity forensic analysis with 98.56% test accuracy and a macro-averaged F1-score of 0.98 across six stroke types. The implementation

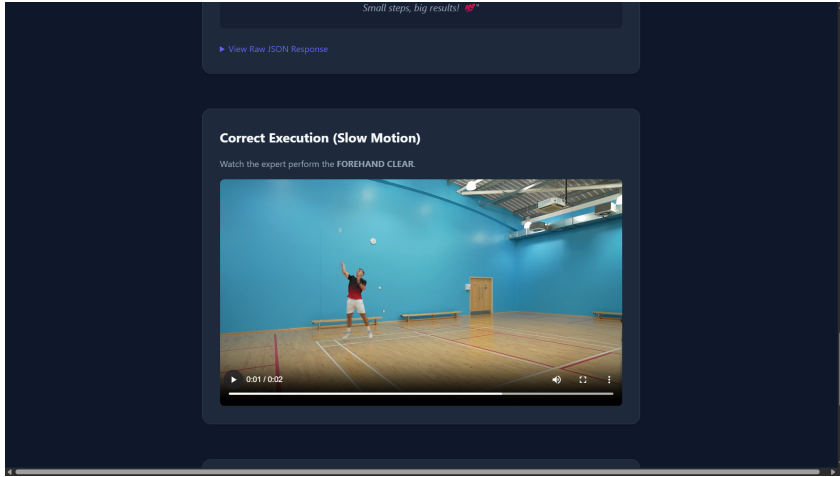


Fig. 12 Correct Execution (Slow Motion) module displaying expert reference footage for the classified Forehand Clear stroke. The slow-motion playback enables players to visually compare their technique against elite biomechanical form.

of a deterministic, configuration-driven spatial cropping strategy successfully eliminated the temporal jitter characteristic of dynamic bounding boxes, ensuring robust and consistent visual stream extraction. The mathematical rigor of the KSI scoring system—employing Sakoe-Chiba constrained DTW, phase-aware attention weighting, 32 hierarchical biomechanical features, and differential kinematics—is seamlessly translated into actionable, skill-appropriate textual drills by an integrated Natural Language Coach. Experimental results demonstrated that Bayesian hyperparameter optimization via Optuna yielded a peak validation accuracy of 98.92%, confirming the capacity of the architecture when optimally configured. The system was validated against existing state-of-the-art methods and demonstrated superior classification accuracy combined with unmatched feedback granularity, providing sophisticated yet accessible biomechanical analysis that existing systems lack.

Future work offers several promising directions to expand the system’s capabilities. Integrating multi-person tracking would extend analysis to doubles matches and crowded environments. A critical area for procedural improvement involves developing a unified, dynamic multi-person tracking mechanism or a self-adjusting bounding box logic that maintains the temporal stability of the current static methodology while actively eliminating the need for manual, per-video configuration overrides. Additionally, deploying quantized models on edge devices like smartphones would democratize access without requiring server-side computation. Incorporating dense surface modeling would also enable deeper analysis of muscle activation and complex torso rotations. Finally, transitioning to data-driven deep metric learning would capture subtler elite motion nuances often missed by hand-engineered features.

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